

Assignment Report on TCP Connection Performance Analysis using Mininet and Wireshark

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1. Introduction

The objective of this assignment was to implement a TCP client-server application in Python and evaluate the performance of two TCP communication methods:

- Persistent TCP Connection
- New TCP Connection for Every Message

The application was executed inside the Mininet network emulator using a bandwidth of 5 Mbps and a network delay of 50 ms. Response time and throughput were measured for message sizes of 128 bytes, 512 bytes, and 1024 bytes. TCP packets were captured and analyzed using Wireshark.

2. Mininet Topology

The default Mininet topology was used.

h1 ----- s1 ----- h2

- h1: TCP Server
- h2: TCP Client
- Switch: s1
- Server IP: 10.0.0.1
- Port: 5000

The topology was verified using the following commands:

nodes
net
pingall

The ping test showed successful communication between the client and the server with 0% packet loss.

3. Explanation of TCP client-server communication.

The TCP server listens on port 5000 and waits for incoming client connections.

The client sends messages in the following format:

MSG_ID|MESSAGE_SIZE|MESSAGE_DATA

Example:

3|512|AAAAAAAAAAAAA...

The server extracts the message ID and size, then replies with an acknowledgement.

Example:

ACK|3|512

The client measures the response time for every message and stores the results in CSV files. The server records all received messages in server_log.txt.

4. Difference between persistent and new-connection modes.

Persistent Connection

In persistent mode, the client creates one TCP connection and sends all messages through the same connection before closing it.

Advantages:

- Lower connection overhead
- Lower response time

- Higher throughput
- Only one TCP handshake

New Connection

In this mode, the client opens a new TCP connection for every message.

Advantages:

- Independent communication
- Simple connection management

Disadvantages:

- Higher overhead
- More TCP handshakes
- Lower throughput
- Increased response time

5. Result table.

The experiment was performed for the following message sizes:

- 128 Bytes
- 512 Bytes
- 1024 Bytes

Each message size was tested in:

- Persistent Mode
- New Connection Mode

Each test sent 10 messages, producing a total of 60 messages.

The following performance metrics were calculated:

- Average Response Time
- Throughput

The results were saved in:

- result_table.csv

roll_no	name	mode	bandwidth_mbps	delay_ms	message_size_bytes	total_messages	average_response_time_seconds	throughput_bytes_per_second	status
CS-BTC24-23	Shahan Uz Zaman	persistent	5	50	128	10	0.20061847110009695	638.026993716523	Success
CS-BTC24-23	Shahan Uz Zaman	persistent	5	50	512	10	0.20055663849993835	2552.894802333643	Success
CS-BTC24-23	Shahan Uz Zaman	persistent	5	50	1024	10	0.2004208703999666	5109.248343031698	Success
CS-BTC24-23	Shahan Uz Zaman	new_connection	5	50	128	10	0.20055152619997899	638.2399696742543	Success
CS-BTC24-23	Shahan Uz Zaman	new_connection	5	50	512	10	0.20044664940000984	2554.2956269538663	Success
CS-BTC24-23	Shahan Uz Zaman	new_connection	5	50	1024	10	0.20034400100007588	5111.208695485782	Success

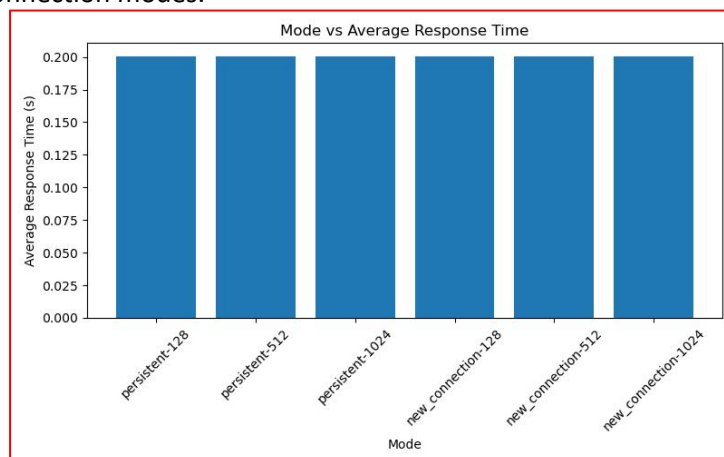
- message_response_log.csv

6. Generated Graphs.

The following graphs were created using Python and Matplotlib:

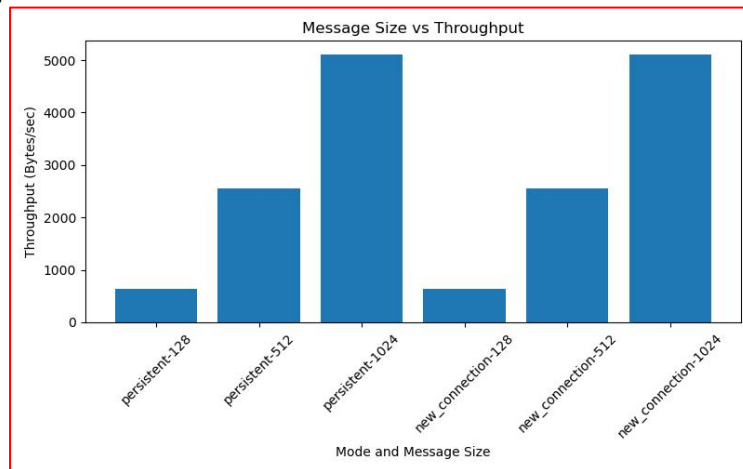
- Mode vs Average Response Time

This graph compares the average response time between Persistent and New Connection modes.



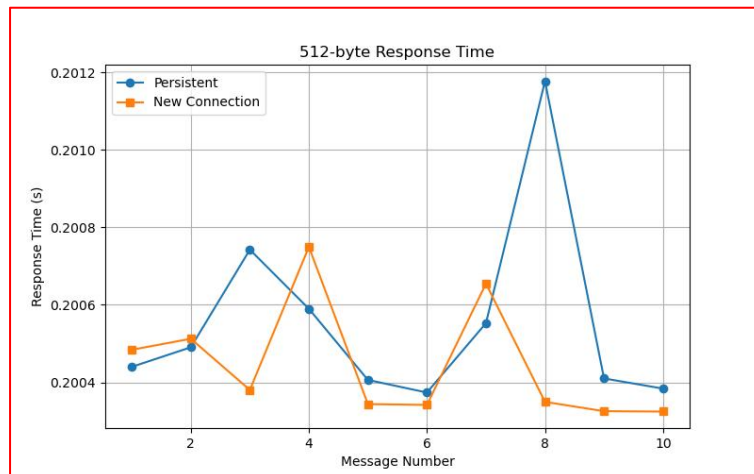
- 2. Message Size vs Throughput

This graph compares throughput for 128-byte, 512-byte, and 1024-byte messages.



- 3. Message Response Time

This graph shows the response time for 512-byte messages in both communication modes.



7. Wireshark Analysis.

Wireshark was used to capture TCP packets with the following display filter:

tcp.port == 5000

The captured packets confirmed the TCP behavior.

Persistent Mode

Observed:

- One TCP three-way handshake
- Multiple data packets
- One connection termination

New Connection Mode

Observed:

- Multiple TCP handshakes
- Multiple TCP connection terminations
- One handshake for every message

These observations match the expected behavior of both communication methods.

8. Answers to Assignment Questions.

A. Why does TCP use a three-way handshake?

The TCP three-way handshake establishes a reliable connection between the client and server. It synchronizes sequence numbers and confirms that both devices are ready to communicate.

B. Which mode had the lower response time?

Persistent mode had the lower response time because the TCP connection is established only once.

C. Why does New Connection mode have more overhead?

Each message requires a new TCP handshake and connection termination, increasing communication overhead and total response time.

D. Which mode is better for CampusChat and why?

Persistent mode is better because chat applications exchange many messages continuously. Keeping the connection open reduces delay and improves performance.

E. Which mode is suitable for NetAttend and why?

New Connection mode is suitable because attendance systems typically send only one request at a time. Opening a new connection for each request is simple and reliable.

F. How does message size affect throughput?

As message size increases, throughput generally improves because more data is transmitted during each communication, reducing the impact of connection overhead.

G. How did Wireshark help verify TCP behavior?

Wireshark allowed visualization of TCP packets, including SYN, SYN-ACK, ACK packets, data transfer, acknowledgements, and connection termination. This confirmed that the implemented client and server followed the expected TCP communication process.

9. Conclusion

This assignment demonstrated how TCP communication behaves under different connection strategies. Persistent connections achieved lower response times and higher throughput because they reused the same TCP connection. New Connection mode introduced additional overhead due to repeated connection establishment and termination. Wireshark successfully verified the TCP handshakes and packet exchanges, while Mininet provided a controlled environment for evaluating network performance.

